SARASWAT ACADEMY

TEST- REAL NUMBERS (CLASS-X)

Note- Each question has 2 marks

Q1: If xy = 180 and HCF (x, y) = 3, then find the LCM (x, y).

Q2: Two numbers are in the ratio 2: 3 and their LCM is 180. What is the HCF of these numbers?

Q3: An army contingent of 678 soldiers is to march behind an army band of 36 members in a Republic Day parade. The two groups are to march in the same number of columns. What is the maximum number of columns they can march?

Q4: Show that $5 + 2\sqrt{7}$ is an irrational number, where $\sqrt{7}$ is given to be an irrational number

Q5: The sum of LCM and HCF of two numbers is 7380.If the LCM of these numbers is 7340 more than their HCF. Find the product of the two numbers.

Q6: If the HCF of 408 and 1032 is expressible in the form 1032 m -408x5, find m.

Q7: Prove that $\sqrt{5}$ is irrational.

Q8: The length, breadth and height of a room are 825 cm, 675 cm and 450 cm respectively. Find the longest tape which can measure the three dimensions of the room exactly.

Q9: A woman wants to organise her birthday party. She was happy on her birthday but there was a problem that she does not want to serve fast food to her guests because she is very health conscious. She as 15 apples and 40 bananas at home and decided to serve them. She wants to distribute fruits among guests. She does not want to discriminate among guests so she decided to distribute equally among all. So

i. How many guests she can invite?

ii. How many apples and banana will each guest get?

Q10: If HCF of two numbers is 15 and their product is 4500, then find how many pairs are possible of those numbers.